

Commodity Return Forecasting

Predicting 1-Week and 1-Month Returns
Using Machine Learning

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1 Introduction

We forecast 5-day and 21-day commodity returns using 60 anonymised features. The methodology is built around preventing look-ahead bias at every stage of the pipeline: target variables are shifted by the forecast horizon to prevent overlap, an embargo gap separates training from testing, position sizing is causal (no future information), and all metrics are computed on non-overlapping periods. We report bootstrap confidence intervals and t -tests on the Information Coefficient to assess statistical significance.

We compare a regularised linear model (ElasticNet) with gradient-boosted trees (XGBoost) trained on both raw and engineered features, and find that the raw-feature XGBoost delivers the strongest risk-adjusted performance.

2 Data Description

The dataset contains $T = 6,549$ daily observations from January 2000 to February 2025. The original 60 features are anonymised. Our targets are the forward returns over h days (in percentage points), where h is either 5 or 21.

The data has many missing values, averaging 43.0%. Since the features have no economic labels, our approach relies entirely on finding statistical patterns rather than using macroeconomic theory.

Table 1: Dataset properties

Parameter	Specification
Period	2000-01-03 to 2025-02-06
Observations (T)	6,549
Raw Features	60
Prediction Horizons (h)	5 days, 21 days
Missing Data Fraction	43.0%
Target Scale	Percentage points

3 Exploratory Data Analysis

3.1 Target Distributions and Stationarity

The return distributions show standard financial characteristics: they are roughly symmetric around zero but have heavy tails (kurtosis of 8.63 for the 1-week horizon and 2.93 for the 1-month horizon). Both return series are strictly stationary based on Augmented Dickey-Fuller tests ($p < 10^{-20}$).

3.2 Missing Data Structure

The dataset exhibits a severe and uneven missing data rate (43.0% overall). Specifically, 27 out of 60 features have over 50% missing values. Feature availability increases dramatically in distinct historical steps (e.g., around 2004 and 2013).

This structural missingness poses a significant machine learning risk: because XGBoost natively handles NaNs, the model might learn to use the *presence or absence* of a feature as a proxy for the temporal regime, leading to subtle temporal overfitting.

3.3 Feature Predictiveness and Stability

While the median feature-target correlation is weak, the right tail of the correlation distribution reveals exceptionally strong individual predictors (correlations up to 0.35 with the 21-day return). In typical financial

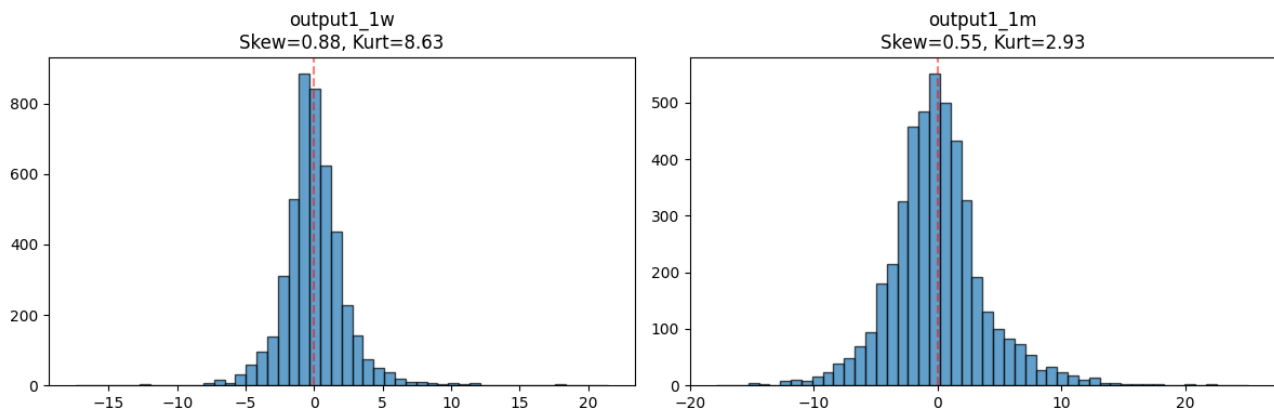


Figure 1: Target distributions for the 5-day and 21-day horizons.

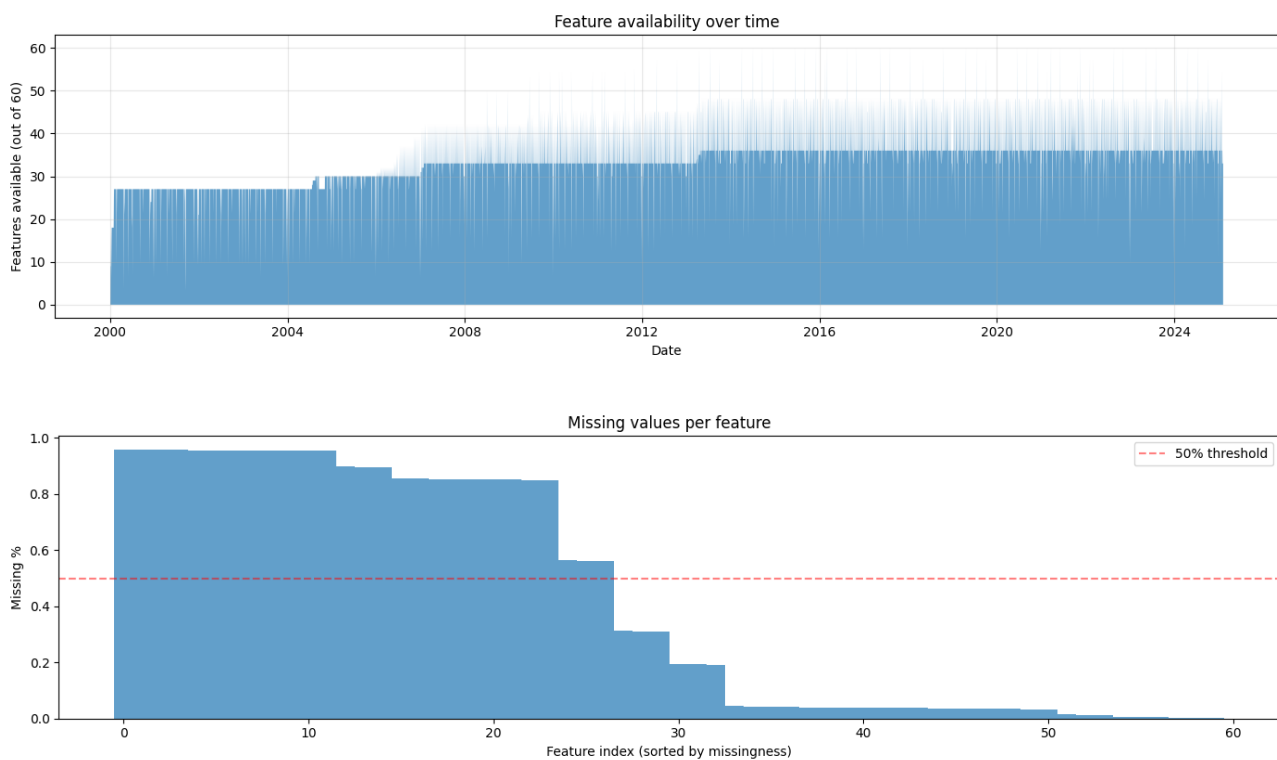


Figure 2: Feature availability over time (top) and missing value percentage per feature (bottom).

datasets, individual feature correlations with forward returns rarely exceed 0.05–0.10. The presence of much stronger correlations raises a question about whether some features are constructed from or correlated with price data in ways that could create upstream leakage. Since the features are anonymised, we cannot verify this, and the reported ICs should be interpreted with this caveat.

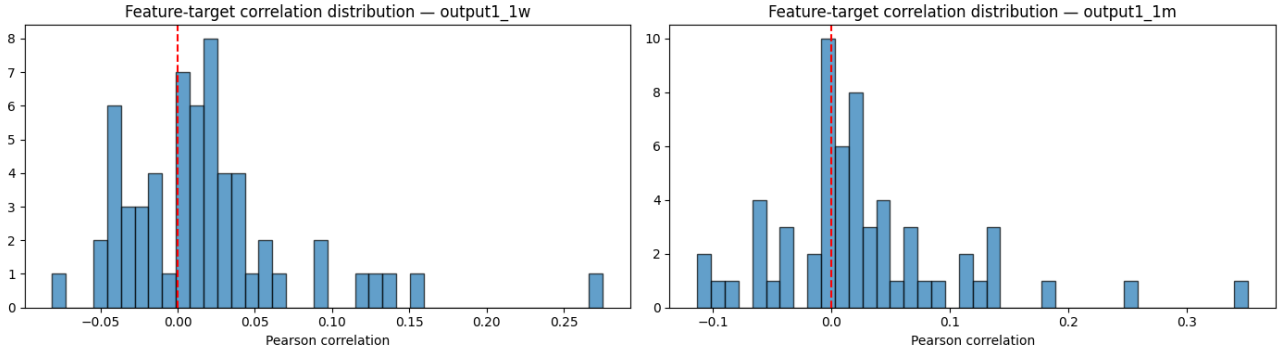


Figure 3: Distribution of Pearson correlations between raw features and targets.

Rolling correlations over 252 days show that the predictive power of these features changes significantly over time. We also found dense clusters of highly correlated features ($|r| > 0.9$). This redundancy requires regularization in linear models and favors tree-based models capable of handling collinear inputs.

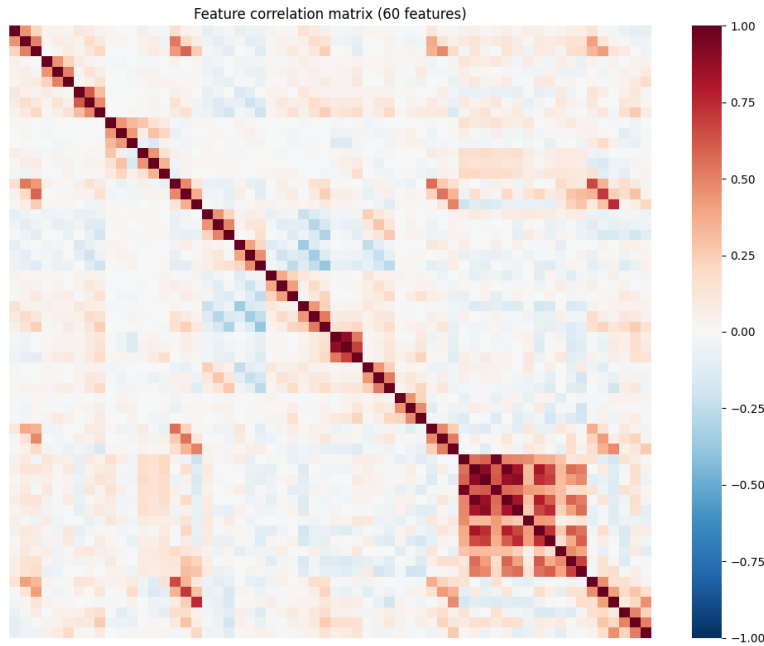


Figure 4: Feature correlation matrix revealing clusters of highly collinear variables.

4 Feature Engineering

We expanded the feature set from 60 to approximately 480 columns using only backward-looking transformations. All transforms are applied uniformly to all 60 raw features — there is no target-based feature selection at this stage (see Section 9 for why this matters).

For a feature x_t at time t , we calculated rolling z-scores to measure how far the current value is from its recent average:

$$z_t^{(w)} = \frac{x_t - \mu_{t-w:t}}{\sigma_{t-w:t} + \epsilon} \quad (1)$$

where $w \in \{21, 63\}$ is the lookback window. Cross-sectional percentile ranks were computed across all features on the same date. This acts as a nonlinear rescaling that squashes outliers, though its interpretation is limited since we only have one asset (it is not a true cross-asset rank).

We measured local stability using 21-day rolling standard deviations of each feature. Feature momentum was calculated as the change over the forecast horizon: $x_t - x_{t-h}$. Finally, we added lagged versions of every feature, shifted by multiples of h to avoid overlap with the prediction window:

$$x_t^{\text{lag}_k} = x_{t-k \cdot h}, \quad k \in \{1, 2\} \quad (2)$$

Target lags are also included as features, with each lag shifted by at least h days to prevent overlap between the return window being predicted and the lagged return being used as input:

$$y_t^{\text{lag}_k} = y_{t-(h+k)}, \quad k \in \{0, 1, 2, 5, 10\} \quad (3)$$

5 Modeling Methodology

5.1 Architectures

We tested three configurations. First, an ElasticNet regression acts as our linear baseline, with mean imputation and standard scaling in a pipeline with internal time-series CV for the L1 ratio. Second, an XGBoost model is trained on the raw 60 features. Third, XGBoost is trained on the full engineered set (~ 480 features). XGBoost naturally handles the 43.0% missing data, so we do not need to fill in missing values manually. We use conservative regularisation: max depth 4, colsample_bytree at 0.5, and L1/L2 penalties.

5.2 Walk-Forward with Embargo

We used 5-fold walk-forward cross-validation. For a given horizon h , the last h observations of each training fold are removed (embargo) to prevent information from the prediction window from leaking into training via target lags or rolling features.

5.3 Early Stopping

Instead of fixing the number of trees by manual tuning (which would be another form of look-ahead), we use early stopping. The last 15% of each training fold (chronologically) is held out as a validation set. Training stops if the validation loss does not improve for 50 rounds. The `n_estimators` parameter (1500) serves only as an upper bound.

6 Backtest Design and Risk Metrics

6.1 Causal Position Sizing

Portfolio weights w_t are sized based on the model's prediction, normalised by the standard deviation of all predictions seen *up to that point* (expanding window):

$$w_t = \text{clip} \left(\frac{\hat{y}_t}{\hat{\sigma}_t^{\text{expanding}}}, -1, 1 \right) \quad (4)$$

where $\hat{\sigma}_t^{\text{expanding}} = \text{std}(\hat{y}_1, \dots, \hat{y}_t)$. For the first 20 predictions, where the expanding standard deviation is unreliable, we use a simple sign-based position (± 1). This replaces a previous version that computed `std(y_pred)` over the entire test fold, which was a mild form of look-ahead.

6.2 NAV and Metrics

Since targets are in percentage points, portfolio returns at each period are $r_t = w_t \cdot y_t / 100$, giving fractional returns suitable for compounding. The NAV evolves multiplicatively:

$$\text{NAV}_t = \prod_{s=1}^t (1 + r_s - c_s) \quad (5)$$

where $c_s = \frac{2 \text{ bps}}{10,000} \cdot |\Delta w_s|$ is the transaction cost at each rebalancing.

All metrics (IC, Hit Rate, Sharpe, Max Drawdown) are calculated on non-overlapping periods. The maximum drawdown is expressed as a percentage of the NAV peak. We report bootstrap 95% confidence intervals (2000 resamples) and a t -test for $\text{IC} \neq 0$ across folds to assess statistical significance.

7 Results

Table 2 shows the IC progression across model configurations. XGBoost on raw features achieves the highest IC on both horizons, outperforming the engineered variant. This result is discussed in Section 9.

Table 2: Mean IC across walk-forward folds (with t -test significance)

Configuration	IC (5d)	p -value	IC (21d)	p -value
ElasticNet	0.402	0.0002	0.241	0.058
XGBoost (Raw)	0.472	0.0001	0.381	0.0008
XGBoost (Engineered)	0.464	0.0001	0.337	0.017

Table 3 and Table 4 show per-fold metrics for the best model (XGBoost on raw features) on each horizon.

Table 3: XGBoost (raw) – 5-day horizon, per-fold out-of-sample metrics

Fold	IC [95% CI]	Hit Rate	Sharpe	Max DD
1	0.504 [0.383, 0.617]	62.1%	2.23	−13.5%
2	0.398 [0.233, 0.538]	61.4%	1.99	−30.7%
3	0.575 [0.419, 0.700]	69.3%	2.99	−14.8%
4	0.466 [0.291, 0.646]	66.7%	3.08	−13.8%
5	0.419 [0.287, 0.589]	69.9%	3.61	−9.4%
Mean	0.472 ± 0.071	65.9%	2.78	−16.4%

Table 4: XGBoost (raw) – 21-day horizon, per-fold out-of-sample metrics

Fold	IC [95% CI]	Hit Rate	Sharpe	Max DD
1	0.400 [0.067, 0.662]	54.1%	0.87	−64.4%
2	0.452 [0.227, 0.726]	67.6%	1.73	−60.3%
3	0.279 [0.060, 0.529]	54.1%	1.00	−53.3%
4	0.485 [0.217, 0.677]	56.8%	1.49	−27.4%
5	0.291 [−0.259, 0.819]	64.9%	1.21	−43.8%
Mean	0.381 ± 0.093	59.5%	1.26	−49.9%

The 5-day model delivers a mean Sharpe ratio of 2.78 with a maximum drawdown of −16.4%, and a statistically significant IC ($t = 14.92$, $p = 0.0001$). The 21-day model also finds a real signal (IC = 0.381,

$p = 0.0008$), but the drawdowns are severe (up to -64.4% per fold), making the signal untradeable without strict risk management.

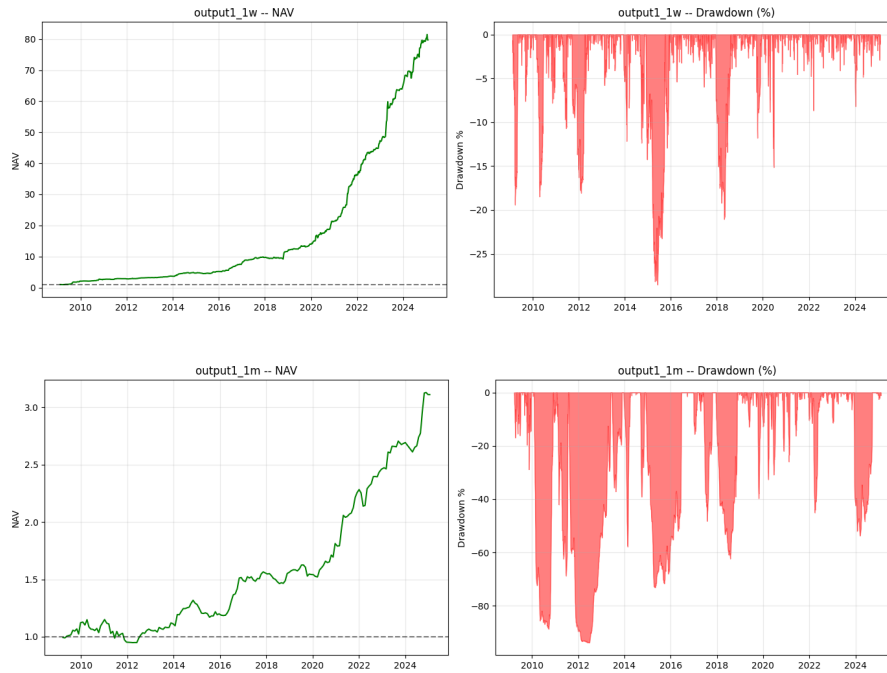


Figure 5: Simulated net performance. The 5-day horizon (top) generates compounding returns with contained drawdowns. The 21-day horizon (bottom) experiences severe variance and systemic drawdowns.

8 Feature Importance Analysis

The XGBoost split gains on the raw-feature model indicate which of the 60 anonymised features drive predictions. Since the features are unlabelled, we cannot provide economic interpretation, but the importance ranking reveals that a small subset of features captures most of the predictive power.

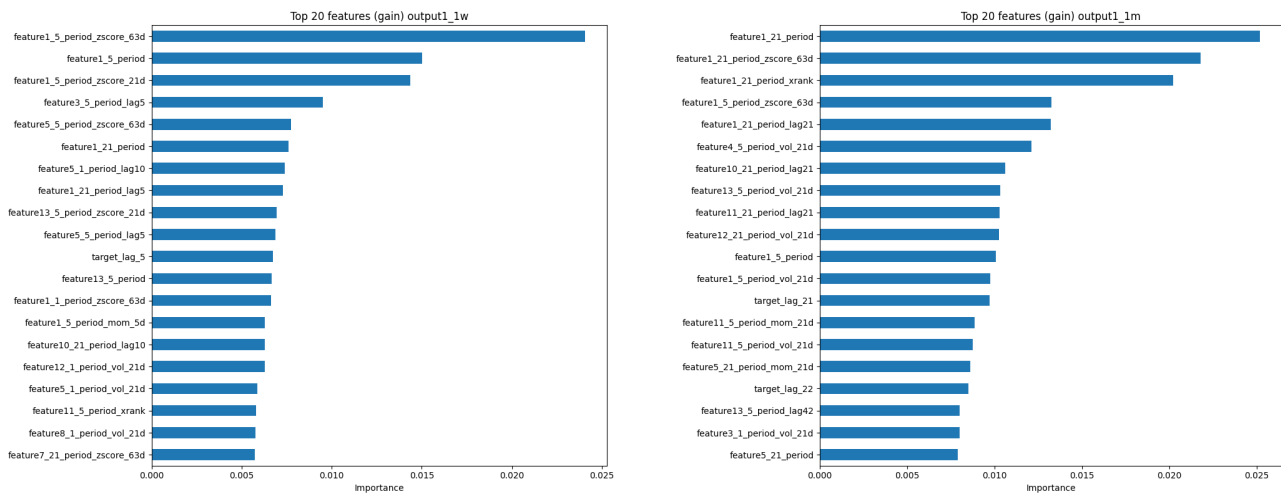


Figure 6: Top 20 features ranked by XGBoost split gain for the 5-day (left) and 21-day (right) horizons.

9 Limitations and Future Work

9.1 Corrected Feature Selection Bias

An earlier version of this pipeline selected which features to enrich (add volatility, lags, and momentum transforms) by computing full-sample correlations between each feature and the target. This is a subtle but real form of look-ahead bias: the selection was conditioned on future test-period target values, even though the transforms themselves were backward-looking. The corrected version applies all transforms uniformly to all 60 features, letting XGBoost's regularisation handle feature selection internally.

After this correction, the engineered features underperform the raw features (IC: 0.464 vs 0.472 on 5d; 0.337 vs 0.381 on 21d). This suggests that applying ~ 420 additional columns without targeted selection introduces more noise than signal. The proper next step would be to perform feature selection inside each walk-forward fold using only training data, which would recover the benefit of enrichment without leakage.

9.2 Suspiciously High ICs

The raw feature correlations with the target (up to 0.35) are unusually high by financial standards. This raises the possibility that the anonymised features contain information derived from or closely related to price data. Without knowing what the features represent, we cannot confirm or rule this out, and the reported ICs should be interpreted with this caveat.

9.3 Single Asset

The entire analysis is on one commodity. We do not know whether these patterns generalise to other assets or whether the model is capturing an idiosyncrasy of this specific instrument.

9.4 Backtest Realism

The 2 basis point cost assumption is optimistic. There is no slippage model, no capacity analysis, and no leverage constraints. The 21-day horizon produces drawdowns exceeding -60% , which would be unacceptable in practice and require volatility targeting or stop-losses, neither of which is implemented here.

9.5 Limited Statistical Power

With 5 walk-forward folds, the t -test on IC has limited power. The 21-day ElasticNet result ($p = 0.058$) is borderline non-significant, and even the significant results should be interpreted cautiously given the small sample of folds.

9.6 Future Work

Potential improvements include implementing volatility targeting and drawdown control for the 21-day signal, performing feature selection inside each fold using only training data (the correct version of the pre-selection approach), testing on multiple commodities, and investigating the data provenance to understand the unusually high feature-target correlations.

10 Conclusion

We find a statistically significant predictive signal in short-term commodity returns using XGBoost. On the 5-day horizon, the model achieves an out-of-sample IC of 0.472 ($p = 0.0001$), a Sharpe ratio of 2.78, and a maximum drawdown of -16.4% . The 21-day signal also exists (IC = 0.381, $p = 0.0008$) but is not viable without risk management given drawdowns exceeding -60% .

The main methodological takeaway is that feature selection must be handled with care. An earlier version that selected features by full-sample target correlation inflated ICs substantially. After correction, raw features outperform the engineered set, confirming that undisciplined feature expansion introduces more

noise than signal. The correct approach is to embed feature selection inside the walk-forward loop, using only training data.

The abnormally high raw-feature correlations remain an open question. Until the data provenance is fully understood, the absolute performance numbers should be treated with appropriate caution.